Wireless Network Setup and Security

Project Network: MySecureWiFi

Introduction

This project demonstrates the design and implementation of a secure wireless network using Cisco Packet Tracer. The network includes both wired and wireless devices connected through a wireless router. Security is enforced using WPA2 authentication and MAC address filtering to prevent unauthorized access.

Network Topology & Hardware

1. Wireless Router (WRT300N): Central device providing connectivity and security
2. Laptops: Authorized wireless devices
3. Unauthorized Laptop: Demonstrates restricted access
4. PC (Wired): Connected via Ethernet

Theory

Wireless Networking: Enables devices to connect without cables using SSID.

WPA2: Secures network with password-based encryption.

MAC Filtering: Allows only selected MAC addresses.

DHCP: Automatically assigns IP addresses.

IP Addressing Table

Device	IP Address
Router	192.168.0.1
PC	DHCP
Laptop 1	DHCP
Laptop 2	DHCP

Implementation

1 Configured SSID and WPA2 password

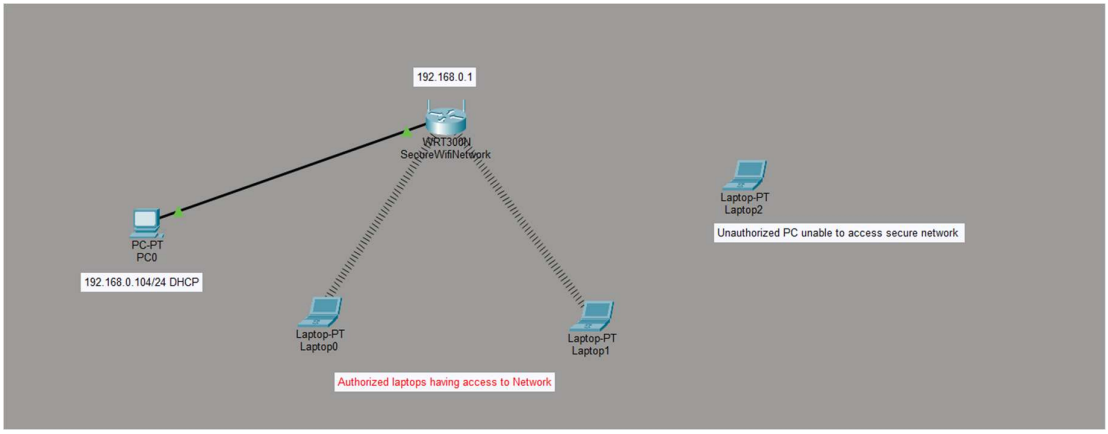
2 Connected authorized devices

3 Enabled MAC filtering

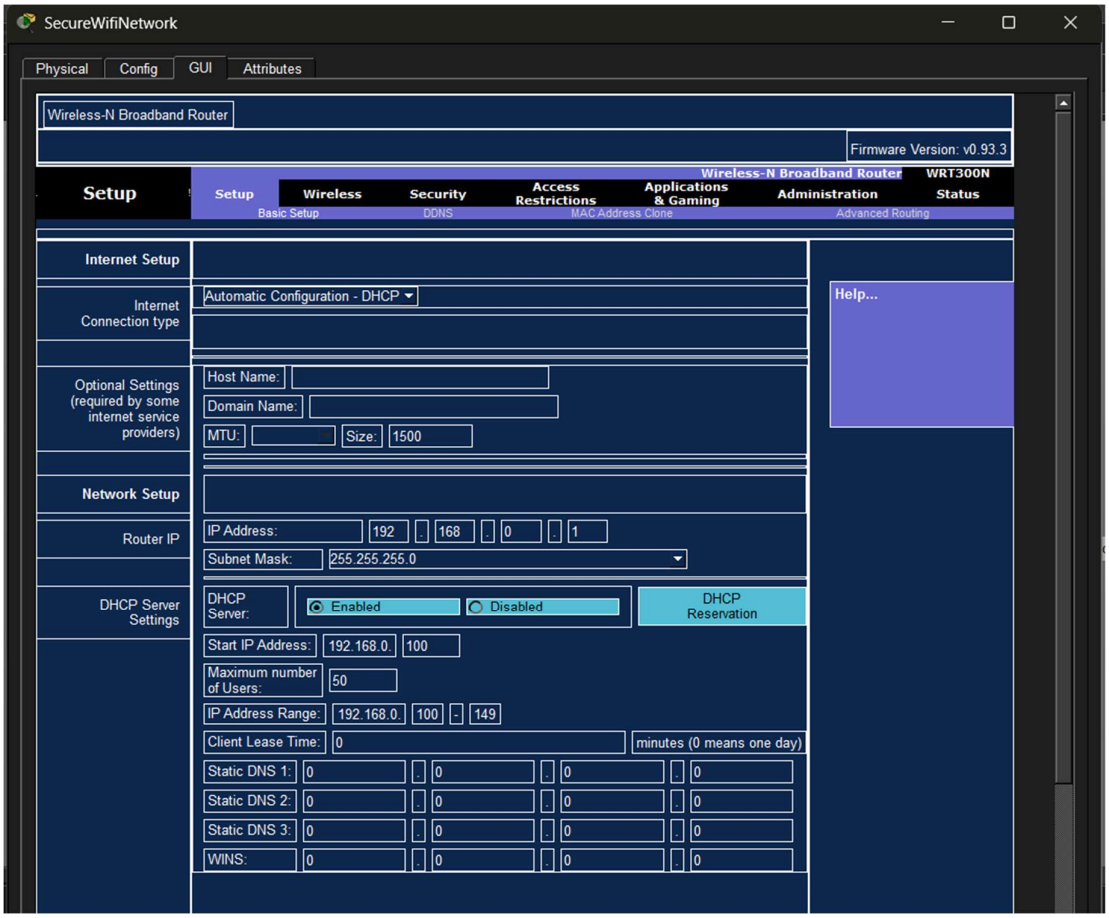
4 Tested network connectivity

Output :-

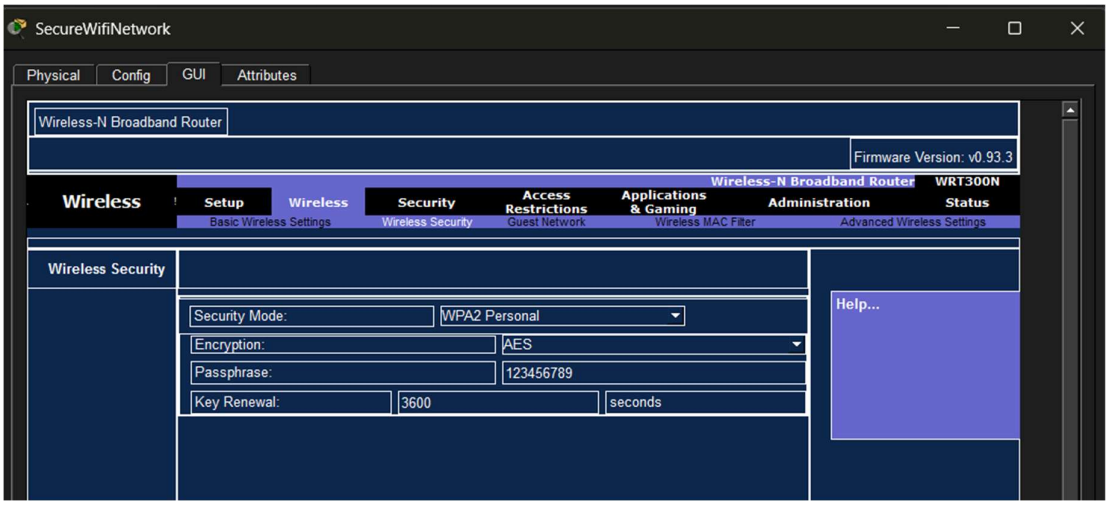
1. Network Topology :



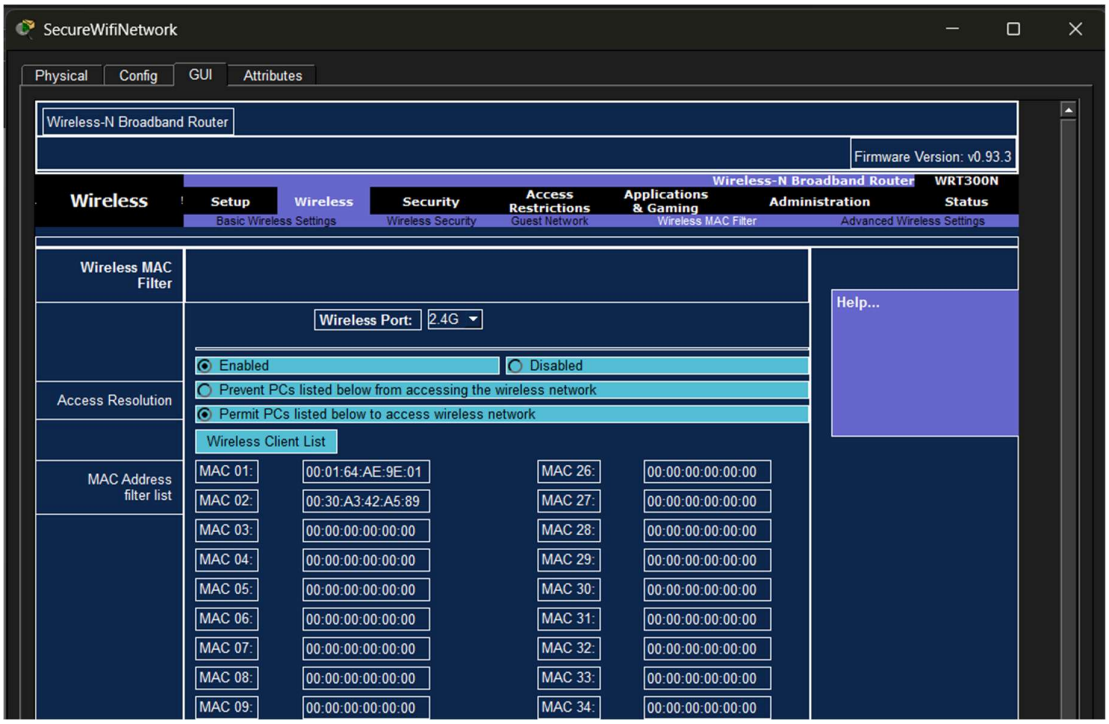
2. Configure Wireless Router :



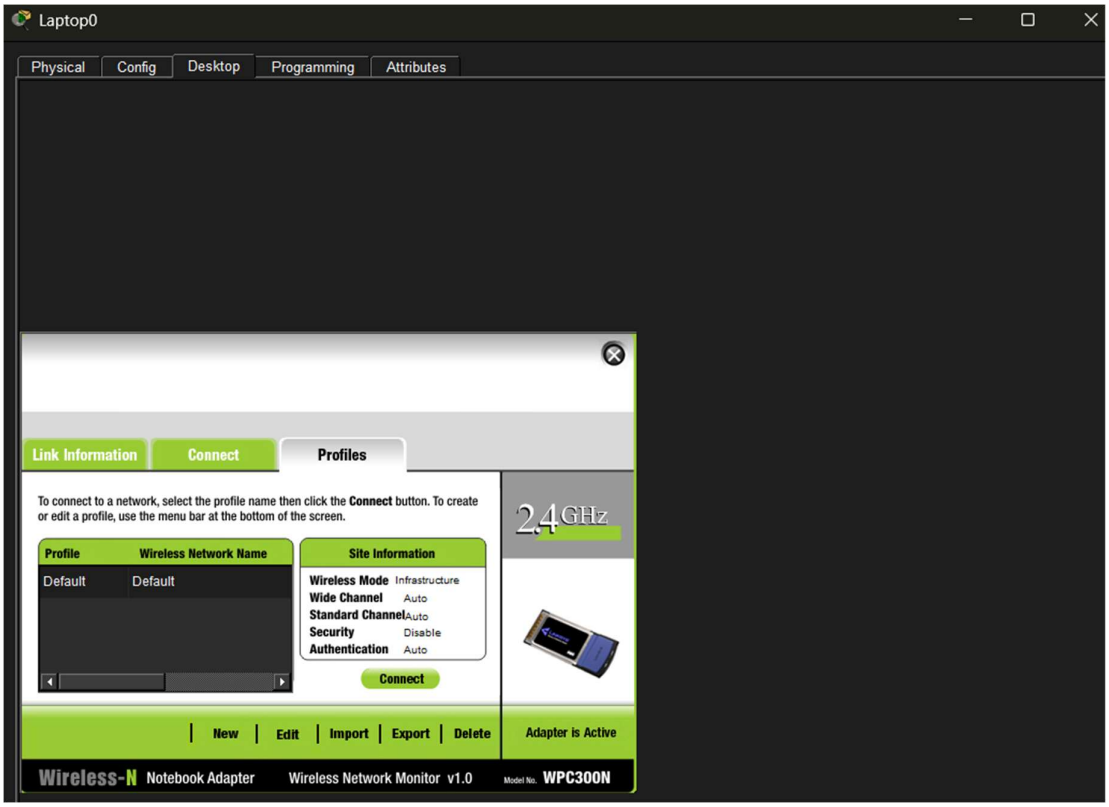
3. WPA2 personal protection :



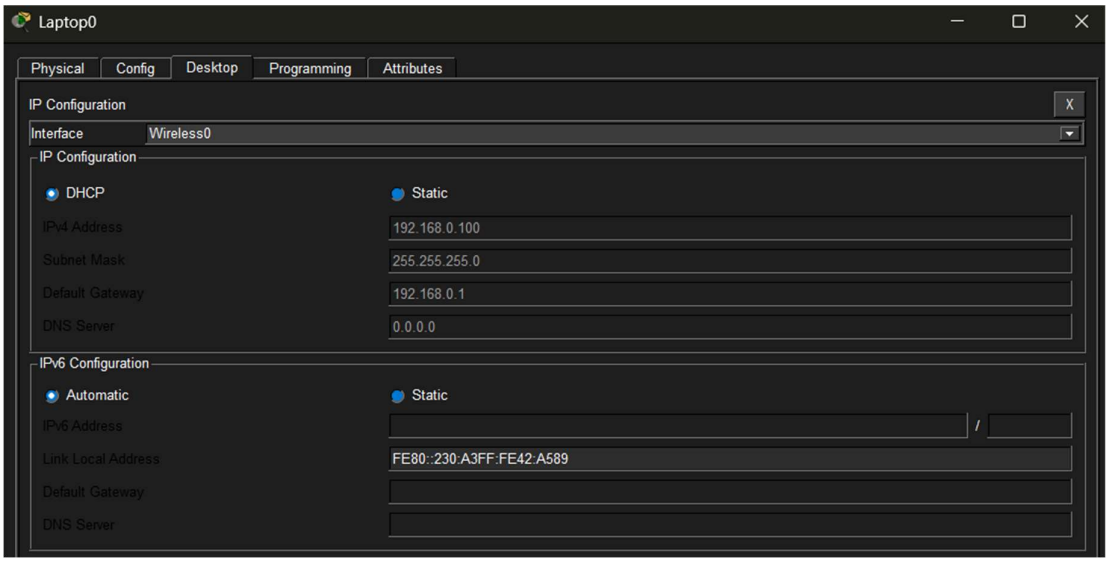
4. MAC Filtering :-



5. Wireless Connection :-



6. Enable DHCP : -



Conclusion

The project successfully demonstrates secure wireless networking using MAC filtering and WPA2.